

Circuit Theory

Chapter 2

Basic Laws

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Basic Laws - Chapter 2

2.1 Ohm's Law.

2.2 Nodes, Branches, and Loops.

2.3 Kirchhoff's Laws.

2.4 Series Resistors and Voltage Division.

2.5 Parallel Resistors and Current Division.

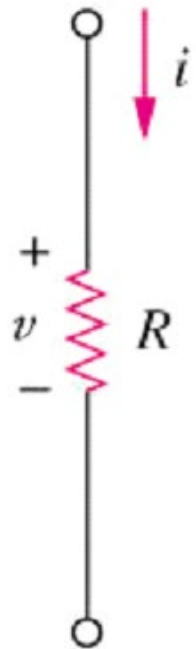
2.6 Wye-Delta Transformations.

2.1 Ohms Law (1)

- Ohm's law states that the voltage across a resistor is directly proportional to the current I flowing through the resistor.
- Mathematical expression for Ohm's Law is as follows:

$$v = iR$$

- Two extreme possible values of R : **0 (zero)** and ∞ (**infinite**) are related with two basic circuit concepts: **short circuit** and **open circuit**.



2.1 Ohms Law (2)

- Conductance is the ability of an element to conduct electric current; it is the reciprocal of resistance R and is measured in mhos or siemens.

$$G = \frac{1}{R} = \frac{i}{v}$$

- The power dissipated by a resistor:

$$p = vi = i^2 R = \frac{v^2}{R}$$

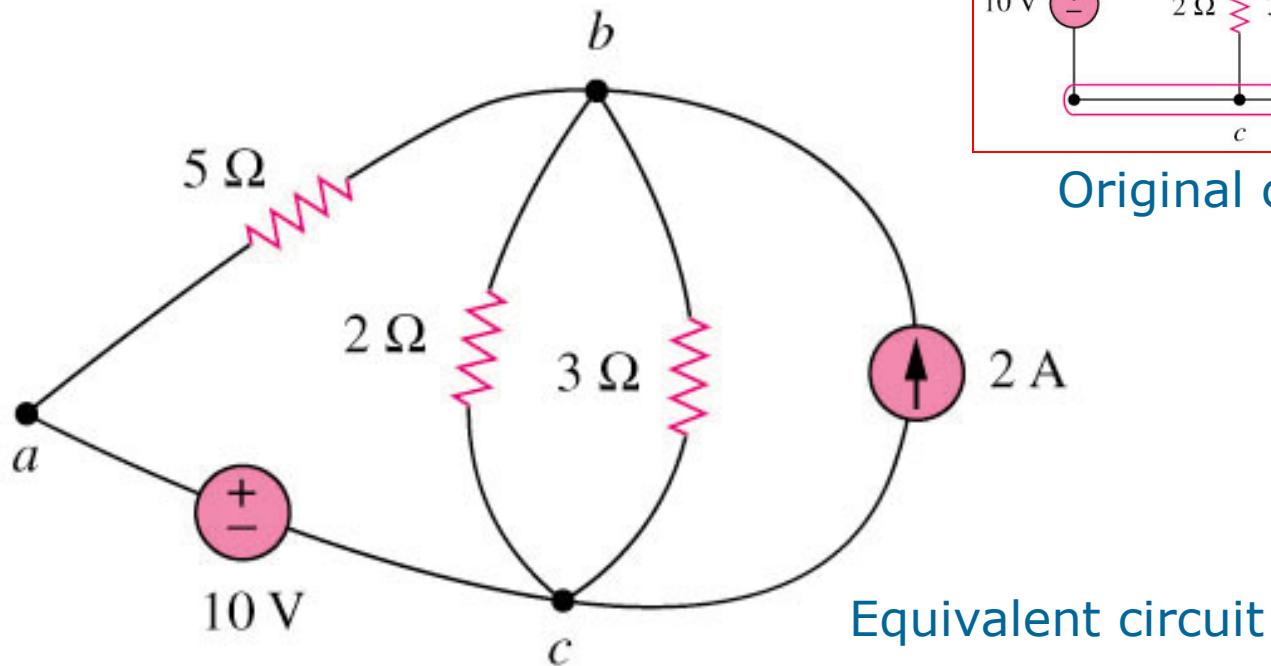
2.2 Nodes, Branches and Loops (1)

- A **branch** represents a single element such as a voltage source or a resistor.
- A **node** is the point of connection between two or more branches.
- A **loop** is any closed path in a circuit.
- A network with b branches, n nodes, and l independent loops will satisfy the fundamental theorem of network topology:

$$b = l + n - 1$$

2.2 Nodes, Branches and Loops (2)

Example 1

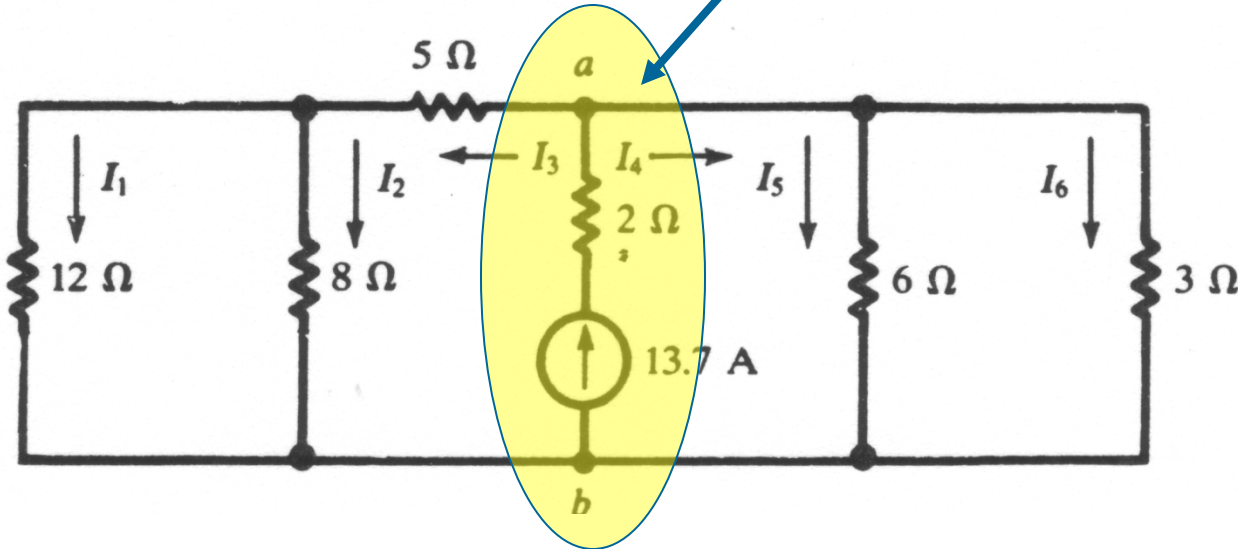


How many branches, nodes and loops are there?

2.2 Nodes, Branches and Loops (3)

Example 2

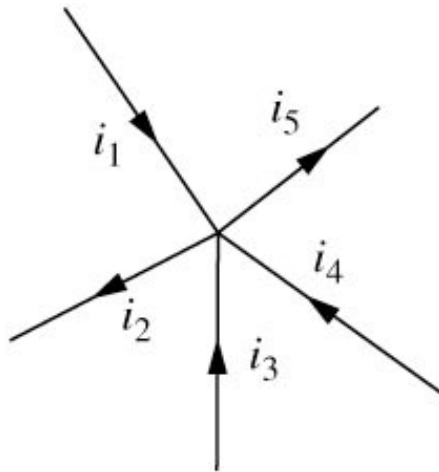
Should we consider it as one branch or two branches?



How many branches, nodes and loops are there?

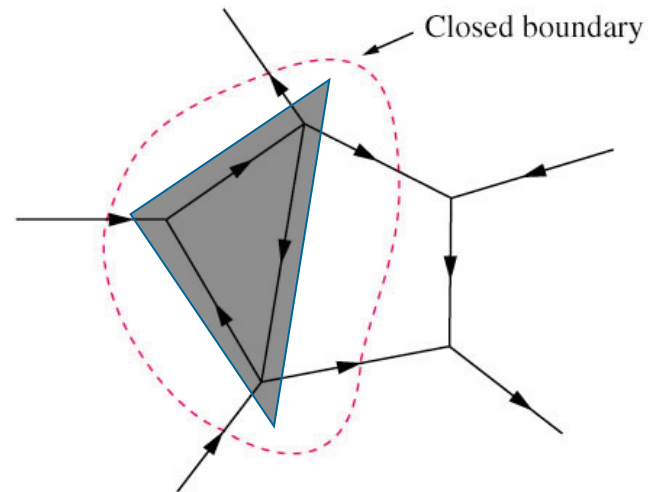
2.3 Kirchhoff's Laws (1)

- Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.



Mathematically,

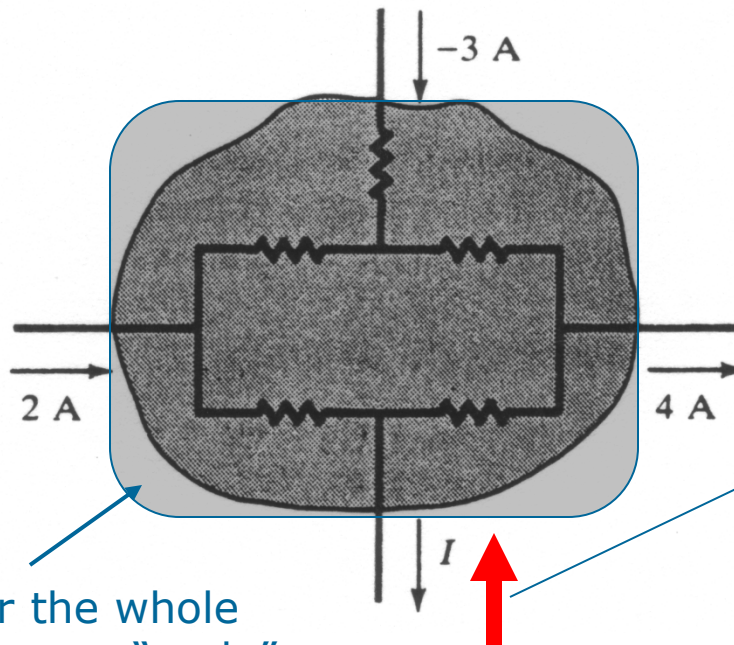
$$\sum_{n=1}^N i_n = 0$$



2.3 Kirchhoff's Laws (2)

Example 4

- Determine the current I for the circuit shown in the figure below.



$$I + 4 - (-3) - 2 = 0$$

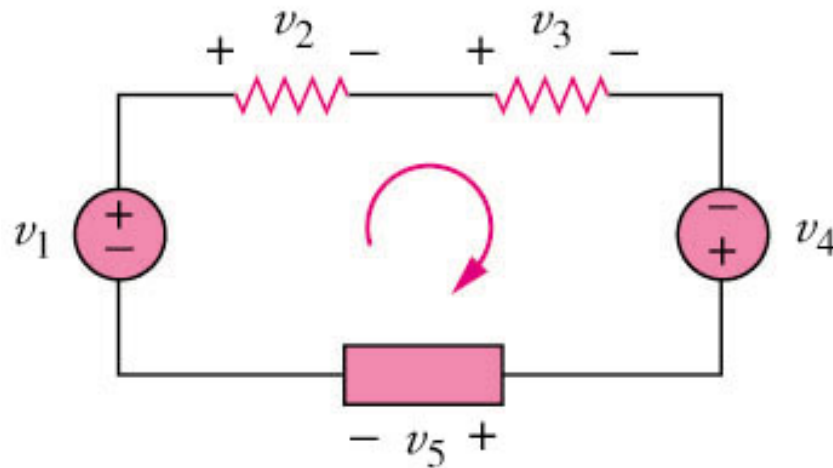
$$\Rightarrow I = -5\text{A}$$

This indicates that the actual current for I is flowing in the opposite direction.

We can consider the whole enclosed area as one "node".

2.3 Kirchhoff's Laws (3)

- Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

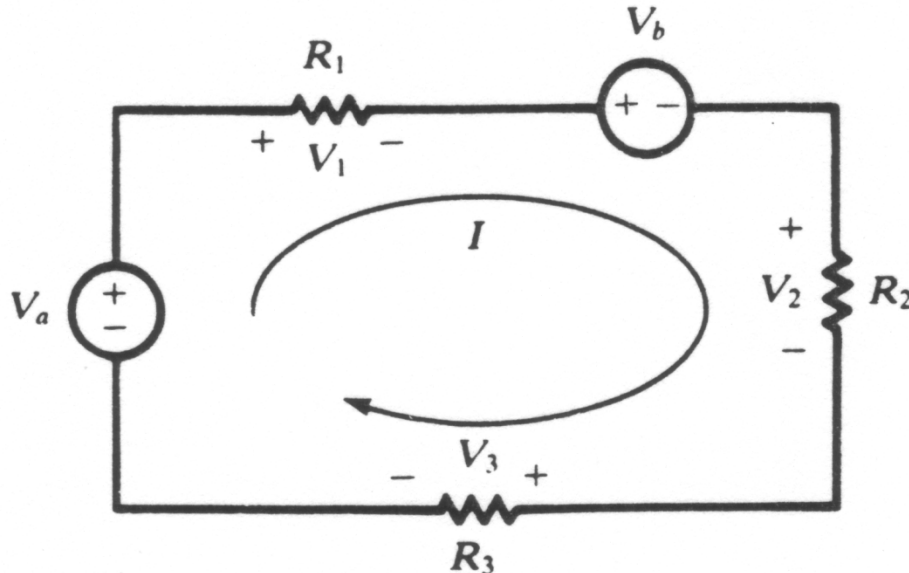


Mathematically,
$$\sum_{n=1}^N v_n = 0$$

2.3 Kirchhoff's Laws (4)

Example 5

- Applying the KVL equation for the circuit of the figure below. Find I.



$$V_a - V_1 - V_b - V_2 - V_3 = 0$$

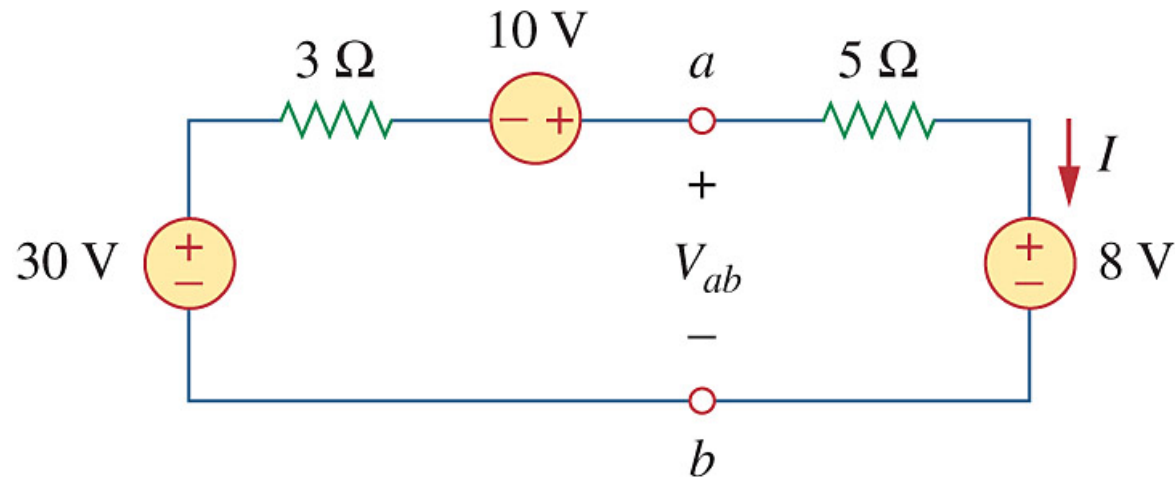
$$V_1 = IR_1 \quad V_2 = IR_2 \quad V_3 = IR_3$$

$$\Rightarrow V_a - V_b = I(R_1 + R_2 + R_3)$$

$$I = \frac{V_a - V_b}{R_1 + R_2 + R_3}$$

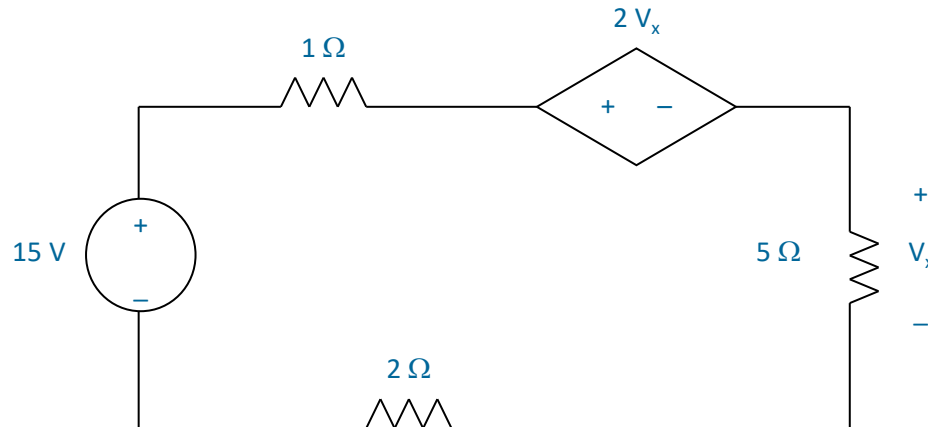
2.3 Kirchhoff's Laws (5)

- Find I and V_{ab} in the circuit.



2.3 Kirchhoff's Laws (6)

- Find V_x in the circuit.



2.4 Series Resistors and Voltage Division (1)

- Series: Two or more elements are in series if they are cascaded or connected sequentially and consequently carry the same current.
- The equivalent resistance of any number of resistors connected in a series is the sum of the individual resistances.

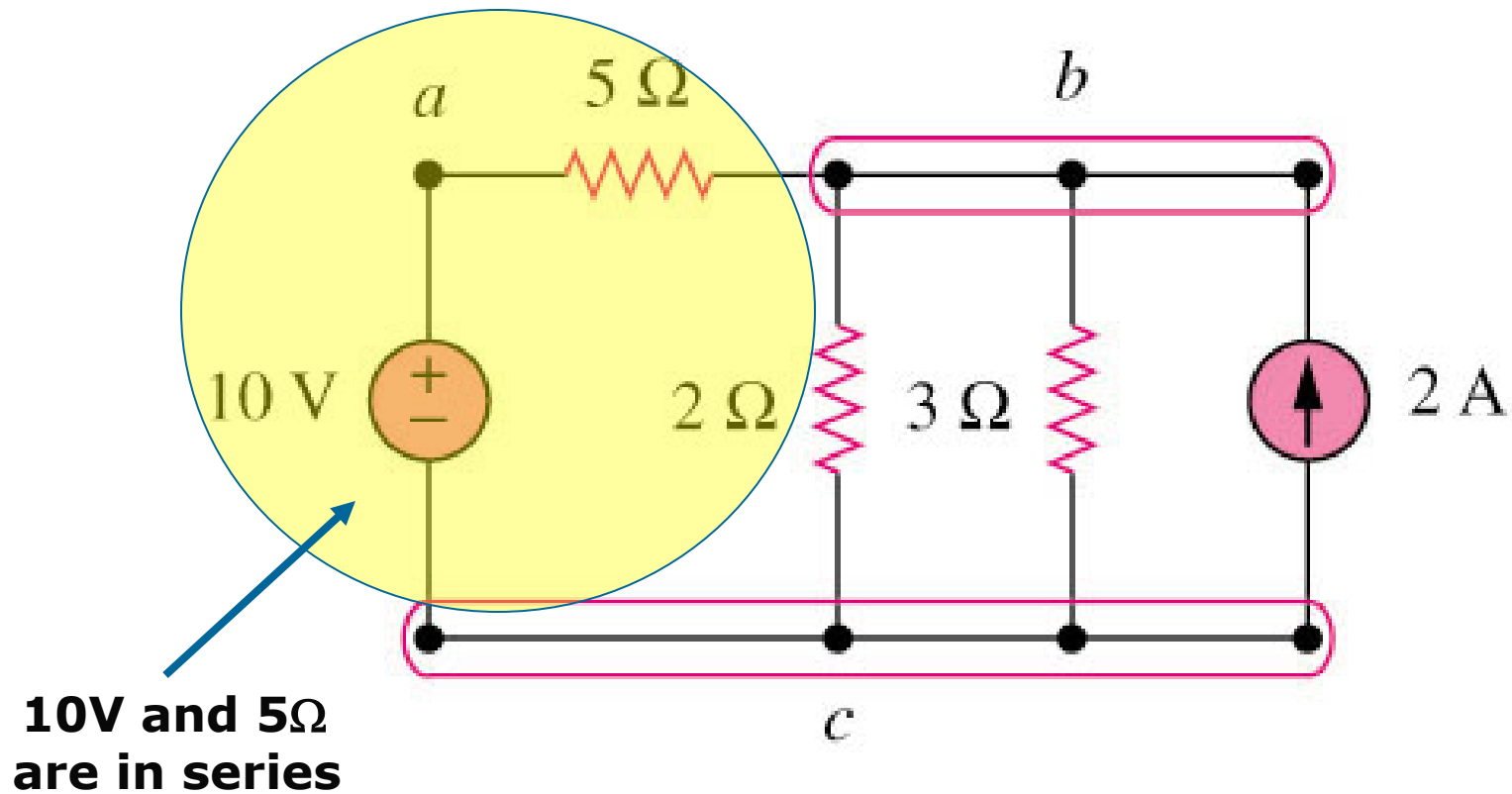
$$R_{eq} = R_1 + R_2 + \cdots + R_N = \sum_{n=1}^N R_n$$

- The voltage divider can be expressed as

$$v_n = \frac{R_n}{R_1 + R_2 + \cdots + R_N} v$$

2.4 Series Resistors and Voltage Division (1)

Example 3



2.5 Parallel Resistors and Current Division (1)

- Parallel: Two or more elements are in parallel if they are connected to the same two nodes and consequently have the same voltage across them.
- The equivalent resistance of a circuit with N resistors in parallel is:

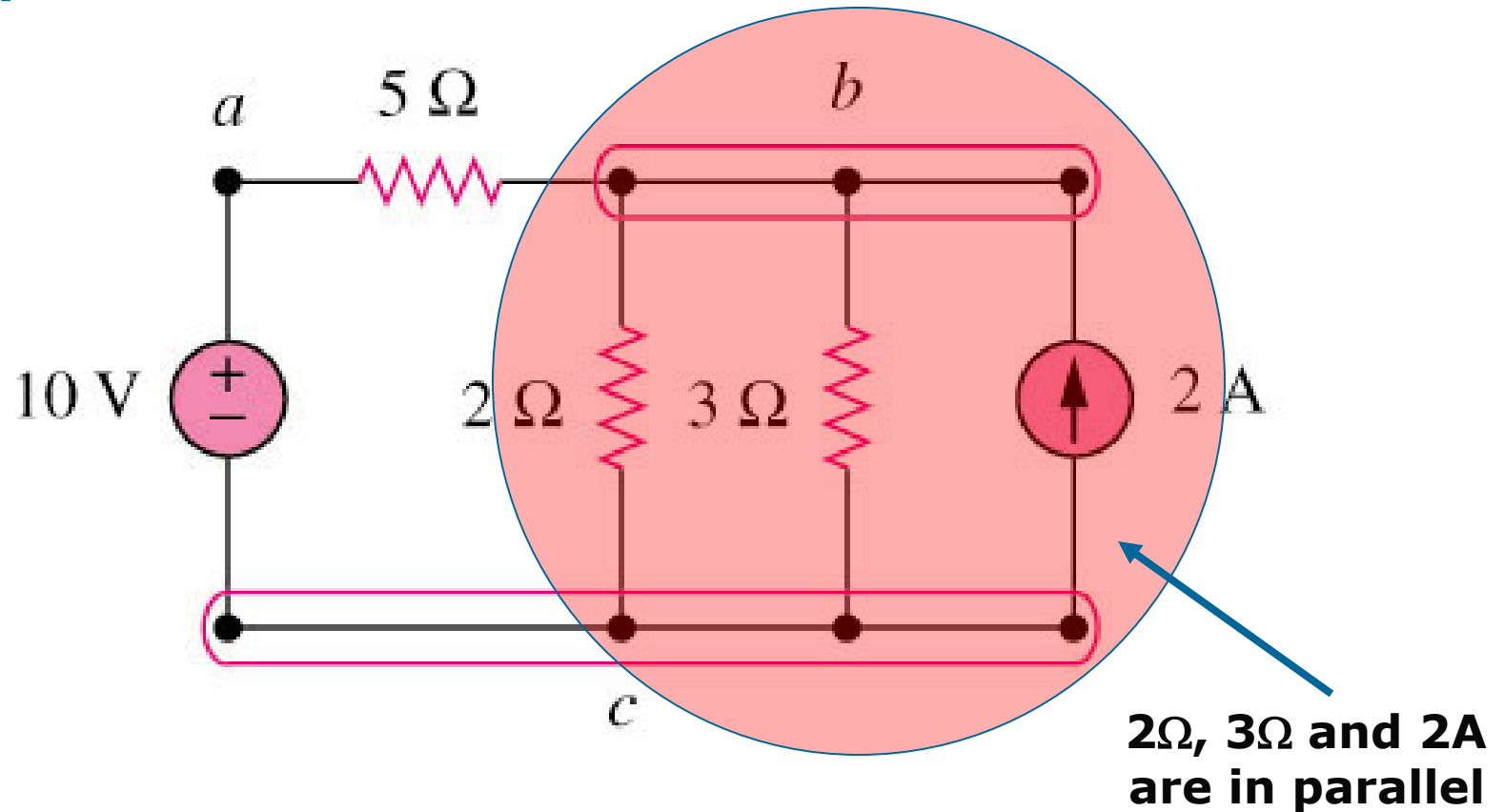
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}$$

- The total current i is shared by the resistors in inverse proportion to their resistances. The current divider can be expressed as:

$$i_n = \frac{v}{R_n} = \frac{iR_{eq}}{R_n}$$

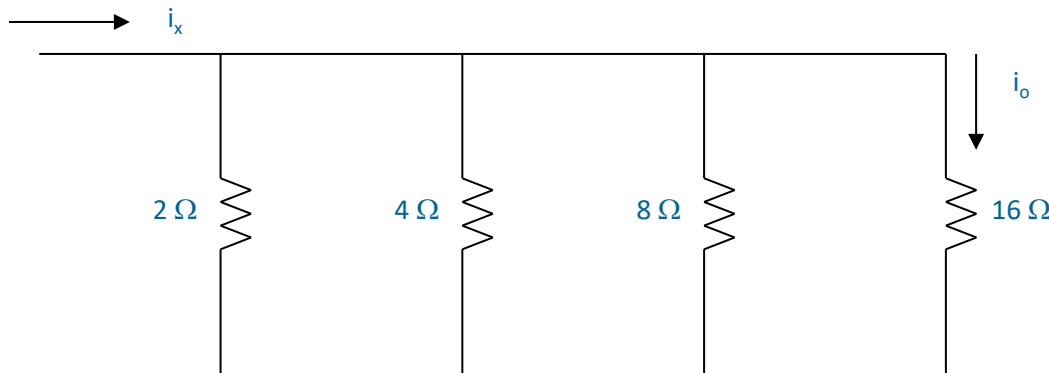
2.5 Parallel Resistors and Current Division (1)

Example 4



2.5 Parallel Resistors and Current Division (2)

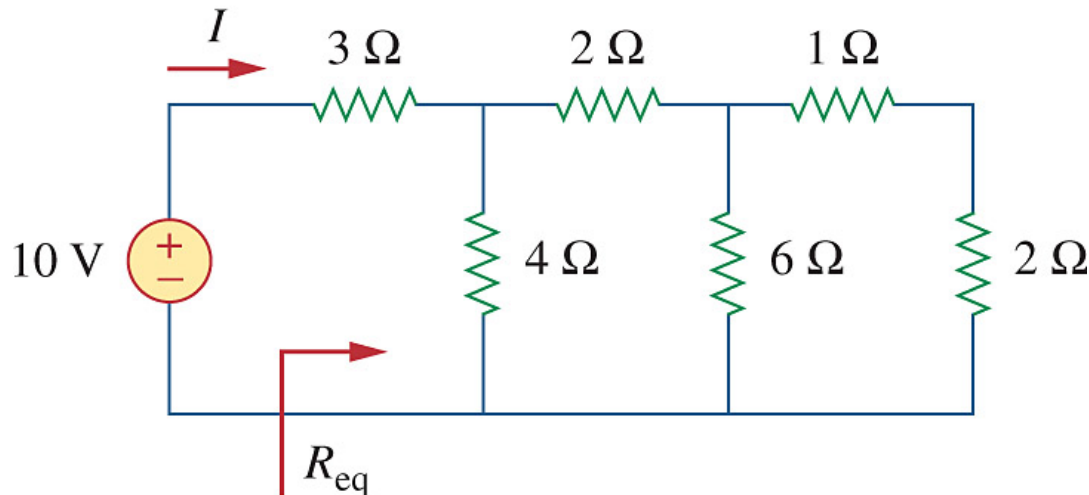
- For the circuit below, $i_o = 2$ A. Calculate i_x and the total power dissipated by the circuit.



Answer: $R_{eq} = 1.0667\ \Omega$, $i_x = 30$ A, $P = 960$ W

2.5 Parallel Resistors and Current Division (3)

- For the ladder network below, find I and R_{eq} .



Answer: $R_{eq} = 5\ \Omega$, $I = 2\text{ A}$

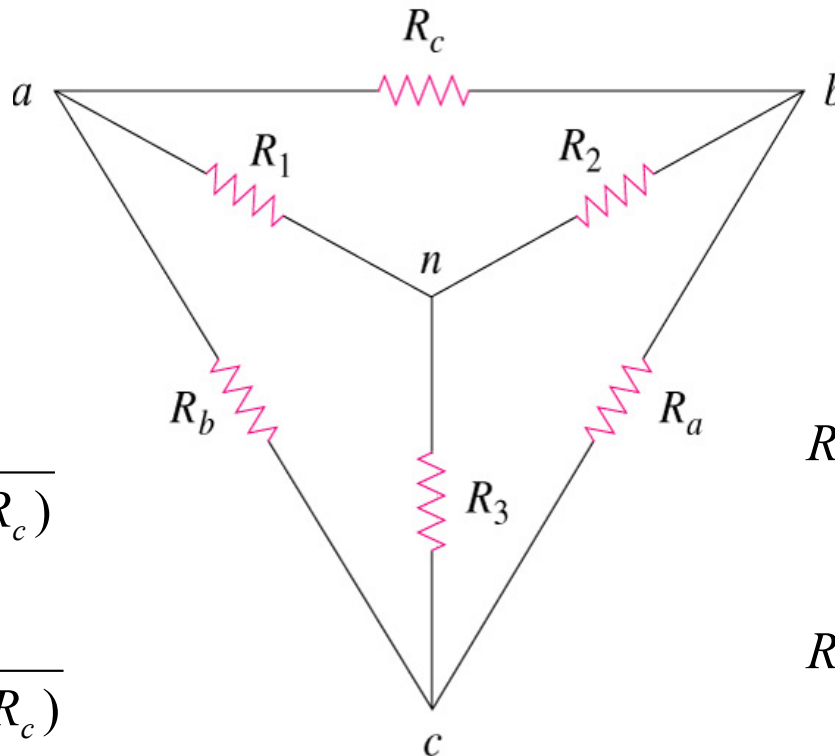
2.6 Wye-Delta Transformations (1)

Delta -> Star

$$R_1 = \frac{R_b R_c}{(R_a + R_b + R_c)}$$

$$R_2 = \frac{R_c R_a}{(R_a + R_b + R_c)}$$

$$R_3 = \frac{R_a R_b}{(R_a + R_b + R_c)}$$



Star -> Delta

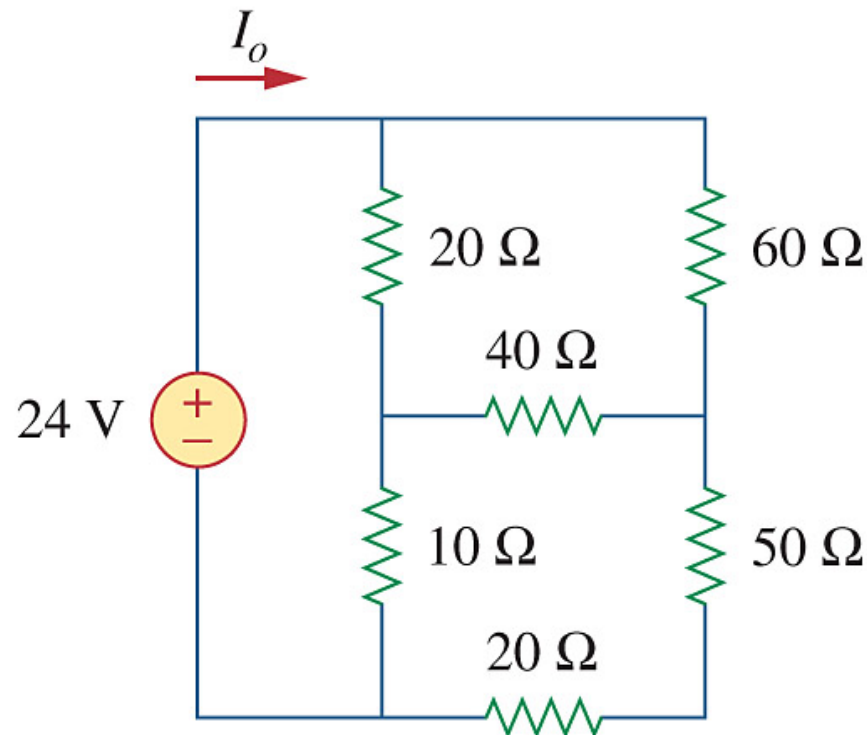
$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

2.6 Wye-Delta Transformations (2)

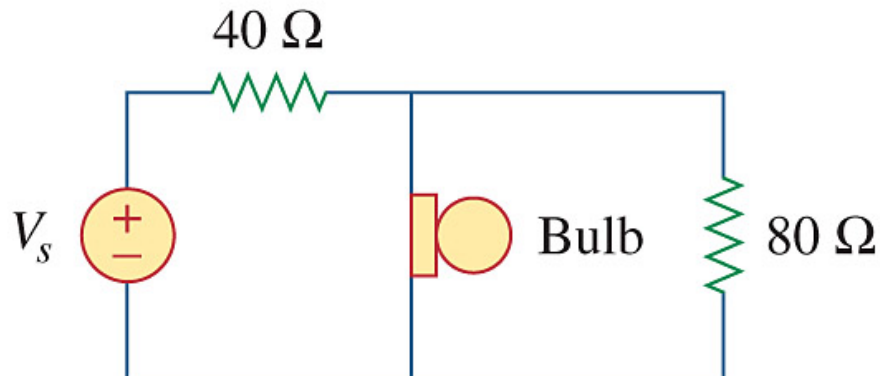
- Calculate I_o in the circuit



Answer: $I_o = 0.997 \text{ A}$

Application Problem

- The lightbulb is rated 120 V, 0.75 A. Calculate V_s to make the lightbulb operate at the rated conditions.



Answer: $V_s = 210\text{ V}$